

Can a classically scale-invariant dark $U(1)$ model explain the NANOGrav signal through a cosmological first-order phase transition?

A pipeline report on the `scale-invariant-dark-u1-zprime` model (`fopt-pta` branch)

DarkAgents

2026-06-10

Abstract

We report an end-to-end automated analysis of a classically scale-invariant (Coleman–Weinberg) gauged dark $U(1)_D$ model with a complex dark scalar Φ , a dark gauge boson Z' , and a vector-like pair of Dirac dark fermions (χ and a mirror partner η), as a candidate explanation of the NANOGrav 15-year (NG15) nano-hertz stochastic gravitational-wave background (SGWB) via a strongly supercooled first-order phase transition (FOPT). The semi-analytic FOPT backend yields a large viable region; the PTA stage finds posteriors that are *interior* to the (user-enlarged) priors, with a maximum a posteriori (MAP) point scoring 14/14 free-spectrum violin bins and parameters ($g_D \simeq 0.59$, $y \simeq 0.38$, $w \simeq 1$ GeV) consistent with the closest literature precedent (arXiv:2502.19478). We conclude that the model *can fit* the NG15 free spectrum in the chosen $w \in [0.05, 10]$ GeV window. However, the cosmological viability of that fit is *conditional and unverified*: the analysis fixed the kinetic-mixing portal $\epsilon = 0$ (secluded dark sector), whereas Standard-Model (SM) reheating and BBN require $\epsilon \sim 10^{-6}$ – 10^{-3} ; a factor- ~ 7 β/H convention mismatch and a saturated GW template ($K_{sw} \rightarrow 1$) weaken the spectral-amplitude interpretation; and no Bayes factor was computed. All five most-decisive cosmological constraints require backend modules that were not run.

1 Prompt

The original user request, verbatim:

“Can a classically scale-invariant $U(1)$ model with a dark scalar, a dark gauge boson and a dark fermion can explain the NANOGrav signal through a cosmological FOPT?”

During the run the user approved each stage and explicitly requested enlarging the PTArcade prior upper bounds to $g_D \leq 2.0$, $y \leq 2.0$, $w \leq 10$ GeV (with the hard cut $y^4 < 1.5 g_D^4$). We report below that the resulting posteriors are interior and the enlarged edges carry no posterior mass.

2 Model description

The model is a classically scale-invariant (no tree-level mass scales; only dimension-4 operators) gauged dark abelian symmetry $U(1)_D$ with coupling g_D , under which the SM is a singlet. Symmetry breaking is radiative (Coleman–Weinberg, CW). The global $U(1)_\chi$ dark-fermion number is conserved, stabilising the lightest dark fermion.

Field content (Table 1). A complex scalar Φ ($q = 1$), whose radial mode is the pseudo-dilaton (scalon) and whose angular mode is eaten by Z' ; the dark gauge boson Z' ; and two Weyl pairs (ψ_L, ψ_R) and (η_L, η_R) forming two degenerate Dirac fermions (χ, η) . The mirror η is introduced purely so that the chiral $U(1)_D$ charges render the sector vector-like and exactly anomaly-free.

Table 1: Field content and $U(1)_D$ charges (from `handoff_model.json`).

Field	Spin	$U(1)_D$ charge	dof	Role
Φ	0	+1	2	complex scalar; radial = scalon, angular eaten
Z'	1	0	3	dark gauge boson, $m_{Z'} = g_D w$
ψ_L	1/2	+1/2	2	Dirac χ (left)
ψ_R	1/2	-1/2	2	Dirac χ (right)
η_L	1/2	-1/2	2	mirror (anomaly partner)
η_R	1/2	+1/2	2	mirror (anomaly partner)

Lagrangian. With $D_\mu \Phi = (\partial_\mu - ig_D Z'_\mu) \Phi$ and $\Phi = \frac{1}{\sqrt{2}} \chi e^{i\theta}$,

$$\mathcal{L} = -\frac{1}{4} F'_{\mu\nu} F'^{\mu\nu} + |D_\mu \Phi|^2 + i \bar{\psi} \not{D} \psi + i \bar{\eta} \not{D} \eta - (y \Phi \bar{\psi}_L \psi_R + y \Phi \bar{\eta}_L \eta_R + \text{h.c.}) - \lambda (\Phi^\dagger \Phi)^2. \quad (1)$$

There is no tree mass term (scale invariance). The Yukawa is gauge invariant ($q_\Phi + q_{\psi_R} - q_{\psi_L} = 0$). Along the flat radial direction the tree potential is $V_{\text{tree}} = (\lambda/4) \chi^4$.

Parameters. The independent parameters are the dark gauge coupling g_D , the dark Yukawa y , and the CW vacuum expectation value $w = \langle \chi \rangle$ (GeV). Dependent quantities follow from CW:

$$\beta_\lambda = \frac{1}{8\pi^2} (3g_D^4 - 8(y/\sqrt{2})^4), \quad m_{Z'} = g_D w, \quad m_\chi = \frac{y w}{\sqrt{2}}, \quad m_{\text{scalon}} = \sqrt{\beta_\lambda} w, \quad (2)$$

with λ fixed by CW minimisation. The critic-agent enforced the existence condition $\beta_\lambda > 0$ as the hard cut $y^4 < 1.5 g_D^4$ (bosonic Z' contribution must dominate the fermionic one), and corrected the vev window from the proposer's $[100, 10^6]$ GeV to $[0.05, 10]$ GeV so that the GW peak can fall in the nHz PTA band (literature benchmarks span 92.9 MeV–10 GeV). The Higgs portal λ_{portal} , the kinetic mixing ϵ , and the renormalisation scale μ are fixed ($\lambda_{\text{portal}} = 0$, $\epsilon = 0$, $\mu = w$, no RG running, no Daisy). The branch-specific feature is that all masses are proportional to w , mapping cleanly onto the single-scalar semi-analytic FOPT backend.

3 Pipeline status

Table 2 summarises the status of every stage. The proposer model was revised by the critic (vev window and $\beta_\lambda > 0$ cut). The FOPT, PTA, constraint and prior-audit stages all completed but carry **warning** status, which propagates to the overall verdict.

4 Preliminary literature search

The librarian verdict is *not novel: a close precedent exists*. The closest match, arXiv:2502.19478 (Balan et al.), is essentially the same model including the Yukawa dark fermion as sub-GeV dark matter, differing by its use of a kinetic-mixing portal. The only distinctive ingredients of the target are the explicit mirror fermion η for anomaly cancellation and the deliberate neglect of kinetic mixing for the FOPT. Table 3 lists the verified papers; all five arXiv identifiers were independently confirmed against arXiv/INSPIRE-HEP during this report.

Table 2: Pipeline stage status (from the corresponding handoff JSONs).

Stage / handoff	Status	Key note
<code>handoff_model</code> (critic)	ok	vev window and $\beta_\lambda > 0$ cut applied
<code>handoff_critique</code>	ok	no red flags; non-novel (closest 2502.19478)
<code>handoff_librarian_preliminary</code>	ok	5 papers; not novel
<code>handoff_fopt</code>	warning	β/H convention (152.7 vs 21.1); α up to 10^{16}
<code>handoff_pta</code>	warning	$K_{sw} \rightarrow 1$ saturation; no Bayes factor
<code>handoff_constraints</code>	warning	$\epsilon = 0$ vs reheating; 5 missing calculations
<code>handoff_prior_audit</code>	warning	22 items; top severity <i>A01</i> (ϵ /reheating)

Table 3: Verified preliminary-literature papers and classification.

arXiv	Title (abbrev.)	Relation to target	Score
2502.19478	Sub-GeV DM and nHz GW from a classically conformal dark sector	same model (adds kinetic mixing)	10
2501.11619	Supercooled PTs in conformal dark sectors explain NANOGrav	structurally similar (scalar+gauge)	9
2501.15649	Supercooled dark scalar PT explanation of NANOGrav	structurally similar (GeV vev)	9
2105.01007	GW from a dark $U(1)_D$ PT in light of NANOGrav 12.5yr	same target, non-conformal	7
2306.17086	NANOGrav signal from a dark conformal PT	background context (strong dynamics)	5

Benchmark hints. The recommended anchor is arXiv:2502.19478 Point A ($g_D = 0.877$, $y = 0.363$, $v = 92.9$ MeV, $T_p = 8.36$ MeV, $T_{\text{reh}} = 12.9$ MeV, $\alpha = 4.99$, $\beta/H = 21.1$), which maps directly onto (g_D, y, w) . Point B ($g_D = 0.79$, $y = 0.401$, $v = 343$ MeV, $\alpha = 62.6$, $\beta/H = 13.3$) gives stronger supercooling. arXiv:2501.15649 BP1–BP4 ($g_D \sim 0.75$ – 0.86 , $w = 0.5$ – 10 GeV, $\alpha \sim 10^2$ – 10^3) bracket the GeV window; arXiv:2501.11619 ($g_D = 0.59$, $\alpha \sim 10^5$) brackets the deep-supercooling end. The librarian flags non-uniform conventions (β/H vs H_*R_* ; T_p vs T_* vs T_{reh} ; CW-fixed vs free λ) that must be reconciled before transferring numbers — a caveat that materialises in the β/H mismatch below.

5 FOPT results

The FOPT stage ran the semi-analytic backend (high- T bounce action, Gaussian percolation, $\mu = w$, no RG/Daisy; scalar and Goldstone contributions neglected, $g_b = 0$) over 374 points, of which 238 are *viable* and 136 failed/excluded. The mapping is $\chi_0 = w$, $g_b(Z') = g_D$ (dof 3), $g_f(\chi) = y/\sqrt{2}$ (dof 8).

The literature anchor (Point A) is reproduced within ~ 10 – 15% on T_p , T_{reh} and α , but the backend reports $\beta/H \simeq 152.7$ versus the literature 21.1 — a factor ~ 7 *convention mismatch* (severity-9 audit item *A02*). Deeply supercooled in-band points reach $\alpha \sim 10^{16}$ (vacuum domination), with $\beta/H \simeq 12$ – 17 and finite, tiny T_p . Table 4 lists the anchor and representative best points. Validation flags (finite values, percolation, eternal-inflation, perturbativity) are all true. The reported $f_{\text{peak}}^{\text{est}}$ is explicitly a template-independent order-of-magnitude redshifted peak, *not* the true spectral peak (which comes only from the PTA-stage template; audit *A21*).

Caveats. The high- T expansion is least reliable exactly in the deep-supercooling regime where the model operates (*A11*); the Gaussian percolation approximation can mis-estimate completion at the extreme- α tail (*A12*); the redshift uses SM $g_*(T)$ although with $\epsilon = 0$ the carrying bath is

Table 4: FOPT benchmarks (from `handoff_fopt.json`); w in GeV, temperatures in GeV.

Point	g_D	y	w	α	β/H	T_{reh}
lit anchor A	0.877	0.363	0.0929	4.7	152.7	1.15×10^{-2}
best_1	0.50	0.20	0.121	7.5×10^{16}	12.3	9.0×10^{-3}
best_3	0.50	0.00	0.707	7.4×10^{15}	12.4	5.3×10^{-2}
best_7	0.54	0.233	0.484	3.6×10^{12}	16.2	3.9×10^{-2}

dark (A18). The GW-spectra overview figure is `fopt_gw_spectra.png`.

6 PTA results

The PTA stage compared the model against the NG15 free spectrum (14 frequency bins) in two modes: a benchmark violin scan and a full PTArcade/ceffyl Bayesian campaign.

Template. The dissipative bulk-flow (*dbf*) template was used (with a bulk-flow *bf* cross-check giving a near-identical peak), justified by the strongly supercooled, vacuum-dominated regime with runaway walls $v_w \rightarrow 1$; the higgsless sound-wave template is invalid for $\alpha \gg 1$. Because $\alpha \gg 1$, the efficiency saturates ($K_{sw} \rightarrow 1$): the amplitude is pinned at its strong-PT ceiling and is effectively insensitive to the true (extreme) α , so the fit constrains essentially β/H and T_{reh} rather than α (audit A03).

Benchmark comparison. Of 945 scanned points, the best score 14/14 violin bins was achieved by 29 interior points. The four ranked best-fit points (Table 5) all score 14/14, with peak frequencies $\sim 10^{-8}$ Hz and $h^2\Omega_{\text{GW}} \sim \text{few} \times 10^{-9}$ at the peak. The benchmark spectrum overlaid on the PTA violins is shown in Fig. 1.

Table 5: PTA best-fit benchmark points (from `handoff_pta.json`); w in GeV.

Rank	g_D	y	w	α	β/H	bins	f_{peak} [Hz]
1	0.507	0.00	1.371	1.6×10^{15}	12.7	14/14	1.90×10^{-8}
2	0.602	0.35	0.500	3.1×10^8	23.6	14/14	1.27×10^{-8}
3	0.686	0.683	2.000	1.4×10^{11}	18.0	14/14	2.96×10^{-8}
4	0.769	0.85	2.828	3.7×10^8	20.7	14/14	2.32×10^{-8}

PTArcade campaign. The ceffyl campaign on NG15 (10^5 samples, acceptance 0.269) used the *user-enlarged* priors $g_D \sim U(0.1, 2.0)$, $y \sim U(0.0, 2.0)$, $\log_{10}(w/\text{GeV}) \sim U(-1.301, 1.0)$, with the hard cut $y^4 < 1.5 g_D^4$. The chains are finite and the posteriors are peaked and *interior*, with no prior-edge contact at 3σ : the enlarged upper edges ($g_D, y \rightarrow 2.0$) carry no posterior mass. The Bayesian estimates are

$$g_D = 0.588 \pm 0.060, \quad y = 0.377 \pm 0.222, \quad \log_{10}(w/\text{GeV}) = -0.002 \pm 0.182 \quad (w \simeq 1 \text{ GeV}), \quad (3)$$

with 68/95/99.7% credible regions all interior. The MAP point ($g_D = 0.535$, $y = 0.294$, $\log_{10} w = 0.097$, i.e. $w = 1.25$ GeV) yields $\alpha = 1.8 \times 10^{15}$, $\beta/H = 13.0$, $T_{\text{reh}} = 98.6$ MeV and scores 14/14 bins. **No Bayes factor / evidence was computed** (`evidence=null`); the “fit” is a goodness-of-region and posterior-shape statement, not a model comparison against SMBHB or other SGWB hypotheses (audit A04). The PTArcade spectrum (benchmark vs MAP, overlaid on violins) is in Fig. 2, and the posterior distributions in Fig. 3.

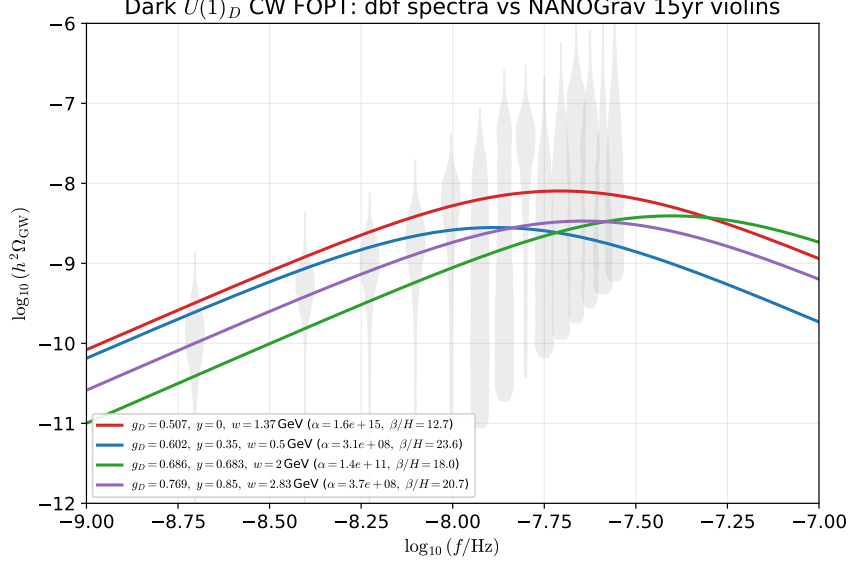


Figure 1: Estimated benchmark GW spectrum vs frequency, overlaid with the NG15 PTA violins. Source: `pta_benchmark_spectrum.pdf`.

7 Constraints

The constraint stage considered 12 constraints (Table 6): 1 directly testable, 2 approximately testable, 3 requiring recast, 5 requiring a new backend, and 1 not applicable. The MAP point is *provisionally allowed but unverified*. The decisive issue is that the FOPT/PTA analyses fixed $\epsilon = 0$, making the dark sector secluded: with strictly $\epsilon = 0$ the latent heat stays dark-only and the model would fail BBN/ N_{eff} . A nonzero kinetic mixing $\epsilon \sim 10^{-6}$ – 10^{-3} is physically required, and that same window triggers the dark-photon laboratory, beam-dump and SN1987A constraints — none of which can be evaluated without Z' decay widths and the portal energy-transfer calculation.

Table 6: Constraints (from `handoff_constraints.json`). Class: D=direct, A=approximate, R=recast, B=new backend, NA=not applicable.

ID	Constraint	Class	Brief	Source
C1	BBN reheating ($T_{\text{reh}} > \text{few MeV}$)	A	SM bath must reheat above ~ 3 – 4 MeV; needs portal transfer	2502.19478
C2	Kinetic-mixing thermalisation	B	ϵ required to thermalise dark with SM	2502.19478
C3	ΔN_{eff} (dark radiation)	B	massless χ ($y=0$), scalon, Z' residue	2502.19478
C4	Z' visible-decay limits	R	BaBar/KLOE/LHCb/NA48/NA64; excludes large ϵ	2603.08430
C5	Beam-dump (E137,CHARM, ν -Cal)	R	low- ϵ long-lived tail	2603.08430
C6	SN1987A cooling	A	light $m_{Z'}$, $\epsilon \sim 10^{-10}$ – 10^{-6}	1611.03864
C7	χ relic / overclosure	B	$\Omega_\chi h^2 \leq 0.12$; secluded annihilation	2502.19478
C8	DM direct detection	B	$\sim \epsilon^2 g_D^2$; depends on f_χ	2603.08430
C9	Scalon decay / BBN injection	B	light scalon, loop-suppressed, possibly long-lived	2502.19478
C10	Perturbativity / $\beta_\lambda > 0$	D	$g_D, y \mathcal{O}(1)$; cut enforced. <i>Satisfied</i> .	internal
C11	Percolation / no eternal inflation	D	checked upstream. <i>Satisfied</i> .	internal (FOPT)
C12	Higgs portal (invisible decay)	NA	$\lambda_{\text{portal}} = 0$	internal

Five missing calculations are flagged as required before any point can be declared allowed: (i)

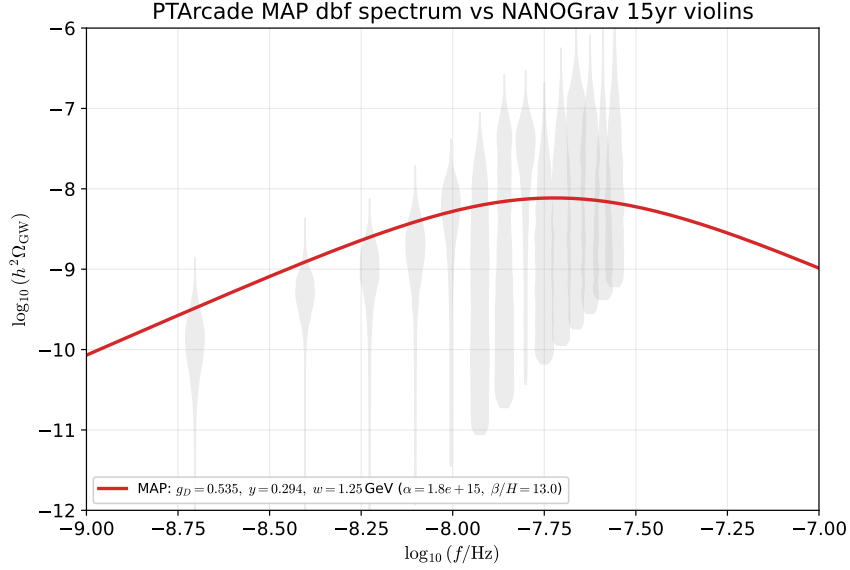


Figure 2: PTArcade Bayesian MAP-point GW spectrum vs frequency, overlaid with the NG15 PTA violins. Source: `pta_ptarcade_spectrum.pdf`.

portal energy transfer / SM reheating temperature; (ii) Z' decay widths/BR/lifetime; (iii) ΔN_{eff} ; (iv) χ relic abundance; (v) scalon decay/lifetime.

8 Assumptions and limitations

The prior-audit ranked 22 items. Table 7 lists the highest-severity ones. The two critical, decision-driving items are A01 (the $\epsilon = 0$ vs reheating tension, severity 10, uncontrolled) and A02 (the β/H convention mismatch, severity 9, uncontrolled). Major limiting items include A03 (template $K_{sw} \rightarrow 1$ saturation, so the fit is α -insensitive), A04 (no Bayes factor), A10 (renormalisation-scale/Daisy systematic), A22 (reheating-viable ϵ may itself be excluded by lab/SN bounds), and A16 (the nHz signal is window-conditional on the critic-chosen $w \in [0.05, 10]$ GeV).

Table 7: Highest-severity assumptions/limitations (from `handoff_prior_audit.json`). Ctrl = controlled.

ID	Description (brief)	Sev	Ctrl	Source
A01	$\epsilon = 0$ secluded; reheating/BBN need $\epsilon \sim 10^{-6}-10^{-3}$	10	no	constraints
A02	β/H convention mismatch (152.7 vs 21.1)	9	no	fopt
A03	dbf $K_{sw} \rightarrow 1$ saturation; fit α -insensitive	8	no	pta
A04	no Bayes factor / evidence computed	8	no	pta
A22	reheating-viable ϵ may be excluded by lab/SN	7	no	constraints
A05	ΔN_{eff} uncomputed (worst at $y \rightarrow 0$)	7	no	constraints
A10	$\mu = w$, no RG, no Daisy systematic on T_p, α, β	7	no	model
A11	high- T bounce action in deep supercooling	6	no	fopt
A13	runaway walls $v_w \rightarrow 1$ assumed	6	no	pta
A16	nHz signal window-conditional on $w \in [0.05, 10]$ GeV	5	yes	critique
A07	enlarged priors ($g_D, y \leq 2.0$); posterior interior	5	yes	pta
A17	non-novel; closest precedent 2502.19478	3	yes	librarian

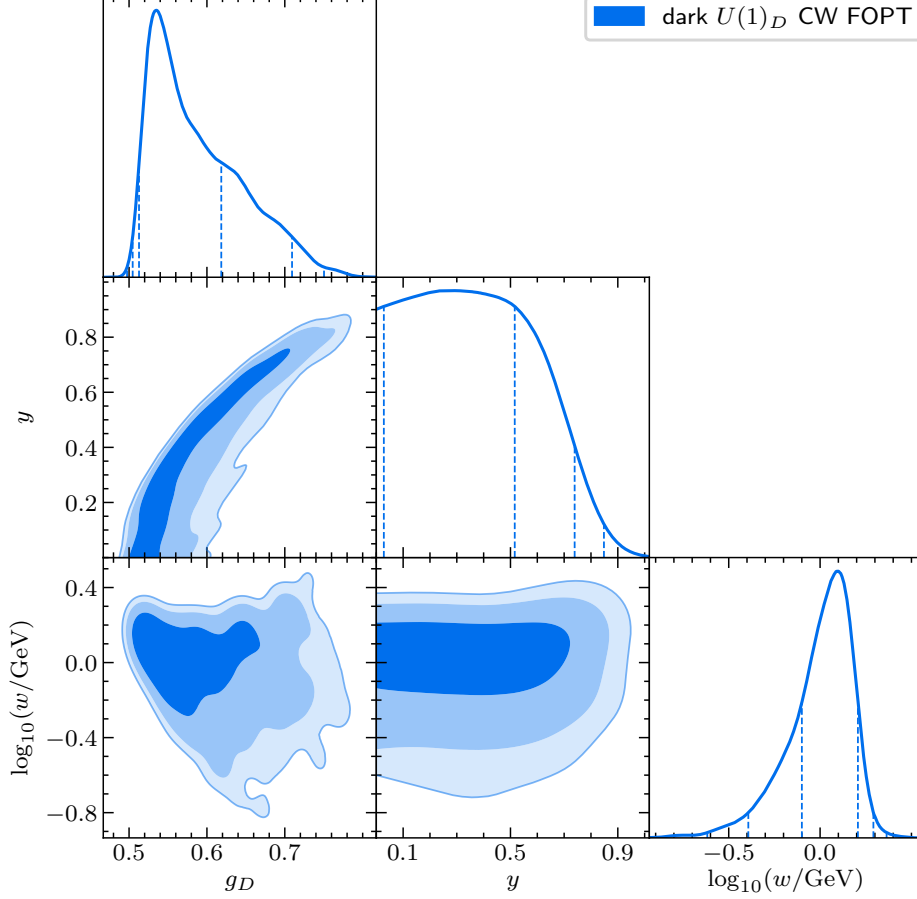


Figure 3: Bayesian posterior distributions on $(g_D, y, \log_{10} w)$ from the PTArcade/ceffyl campaign on NG15. Posteriors are peaked and interior; enlarged prior edges carry no mass. Source: `pta_posteriors.pdf`.

9 Inconsistencies and reference caveats

One reference-content inconsistency was identified during citation validation:

- The PTA and prior-audit handoffs describe arXiv:2511.15687 as the source of the “dbf bulk-flow” GW template and as an “expansion-corrected supercooled template”. The arXiv identifier is *valid and exists*, but the paper is actually “*Impact of cosmic expansion on gravitational wave spectra from strongly supercooled first-order phase transitions*” by M. Lewicki and V. Vaskonen — a paper on cosmic-expansion corrections to supercooled-PT spectra, *not* a dedicated dissipative-bulk-flow template paper. We therefore cite arXiv:2511.15687 only for its actual subject (cosmic-expansion corrections / cross-check context) and flag that the “dbf template” attribution in the upstream handoffs is not corroborated by the cited paper’s content. The numerical PTA results are unaffected (the template was implemented in the backend code), but the bibliographic attribution should be treated with caution.

No other numerical conflicts between handoffs were found; the FOPT β/H mismatch is an internal-convention issue (reported as A02), not a cross-handoff data conflict.

10 Conclusions

Can the model explain NANOGrav through a FOPT? *For the signal fit: yes, conditionally.* Within the critic-chosen $w \in [0.05, 10]$ GeV window, the strongly supercooled FOPT produces a nHz SGWB that fits the NG15 free spectrum: the MAP point and the four ranked benchmarks each cover 14/14 violin bins, the Bayesian posteriors are peaked and interior (no enlarged-prior-edge mass), and the preferred parameters ($g_D \simeq 0.59$, $y \simeq 0.38$, $w \simeq 1$ GeV) are consistent with the closest literature precedent (arXiv:2502.19478).

For cosmological viability: not established. The result is conditional and partly uncontrolled. (1) The whole chain fixed $\epsilon = 0$; SM reheating and BBN require $\epsilon \sim 10^{-6}$ – 10^{-3} , and that window also triggers dark-photon lab/SN bounds that were not evaluated (A01, A22, C1–C6). (2) A factor- ~ 7 β/H convention mismatch (A02) and the saturated $K_{sw} \rightarrow 1$ template (A03) mean the fit effectively constrains β/H and T_{reh} but not α , and may be systematically biased. (3) No Bayes factor was computed (A04), so “explains” is a goodness-of-region statement, not a model-comparison result against SMBHB. (4) Five decisive cosmological observables remain uncomputed.

Robust conclusions: the model is anomaly-free and self-consistent; couplings are perturbative and $\beta_\lambda > 0$ throughout (C10); the transition percolates without eternal inflation (C11); the GW peak lands in the PTA band for the chosen w window; and the PTA fit quality is high and interior. *Conditional/blocked:* cosmological viability, pending a kinetic-mixing portal, reheating, ΔN_{eff} , relic-abundance, scalon-decay, and β/H -reconciliation calculations, plus an evidence/Bayes-factor determination.

References

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